Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *MMLU* | ***Context:*** *South Asian Agricultural and Environmental Science — Mymensingh–Telangana–Andhra Pradesh Deployment*

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The paper is silent on annotator demographics, selection criteria, regional or disciplinary backgrounds, or inter-annotator agreement [Q19]. The only documented human reference is US Amazon Mechanical Turk workers (34.5% accuracy) [Q22], with expert-level estimated from 'real-world test-taker human accuracy at the 95th percentile' on US standardized exams. The HuggingFace dataset card tags annotations_creators as 'no-annotation' [DATASET Concern 8]. The authorship team is entirely US-based [Q10] and content is sourced from US/Anglophone institutions [Q19, Q77]. There is no evidence — and no plausible mechanism — for South Asian agronomic, Mymensingh-dialect, or Telugu-region expert validation of any answer key. The user explicitly confirmed [elicitation A3] that cross-border water/agricultural questions could yield divergent correct answers and that practice differences between regions are not reflected in answer keys. Empirical sampling confirmed multiple items where US-specific facts (Social Security age increases, US birth control prevalence, US 2020 moral norms) are encoded as universally correct [DATASET-D1, D9, D10, D11, D12]. OC is HIGH-priority and represents a fundamental convergent and external validity violation.

ID	Evidence
Q10	"Dan Hendrycks, UC Berkeley; Collin Burns, Columbia University; Steven Basart, UChicago; Andy Zou, UC Berkeley; Mantas Mazeika, UIUC; Dawn Song, UC Berkeley; Jacob Steinhardt, UC Berkeley" (p.1)
Q19	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)
Q22	"Human-level accuracy on this test varies. Unspecialized humans from Amazon Mechanical Turk obtain 34.5% accuracy on this test. Meanwhile, expert-level performance can be far higher. For example, real-world test-taker human accuracy at the 95th percentile is around 87% for US Medical Licensing Examinations, and these questions make up our "Professional Medicine" task. If we take the 95th percentile human test-taker accuracy for exams that build up our test, and if we make an educated guess when such information is unavailable, we then estimate that expert-level accuracy is approximately 89.8%." (p.3)
Q77	"We also had RoBERTa continue pretraining on approximately 1.6 million legal case summaries using Harvard's Law Library case law corpus case.law, but after fine-tuning it only attained 36.1% accuracy." (p.8)
D1	For which of these two scenarios does the main character (who uses I/me/my) do something clearly morally wrong, **according to ordinary moral standards in the US as of 2020**? — Question explicitly frames moral standards as US-specific, 2020 vintage
D9	The age at which one can receive full benefits from social security" with answer "Has been increasing — US Social Security system — entirely US institutional context
D10	Which of the following is the most common method of birth control for married couples in the **U.S.** — Explicitly US demographic data
D11	From 1988 to 1990 among heterosexuals in the **US**, the number of unmarried adults aged 20 to 45 who report having multiple partners has: declined for both men and women — US-specific behavioral survey data
D12	"For which of these two scenarios does the main character ... do something clearly morally wrong, according to ordinary moral standards in the US as of 2020?" (repeated framing across all 5 examples) — Every moral_scenarios question frames correctness relative to US 2020 norms

Strengths

- Authors commit to publishing question sources to enable downstream auditing [Q112]

- Memorization analysis suggests labels are not artifacts of pretraining contamination [Q107, Q108], so label correctness is at least an honest measurement of US-anchored ground truth

ID	Evidence
Q107	“However, in Figure 13, we see that accuracy and question entropy are not positively correlated, suggesting that the test's low-entropy questions do not correspond to memorized (and thereby correctly predicted) answers.” (p.14)
Q108	“This suggests that our exact questions were not memorized.” (p.14)
Q112	“To reduce the probability that future models encounter exact questions during test-time, we will provide a list of question sources.” (p.14)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground truth labels reflect US/Anglophone academic standards [Q19, Q22], not Bangladeshi or Telugu-region stakeholder perspectives. The user [elicitation A3] confirms they would not be considered authoritative for cross-border or Bangladeshi-context agricultural questions.

OC-2: Substantial disagreement is expected for: (a) moral_scenarios items (US 2020 framing) [DATASET-D1, D12]; (b) any cross-border water-policy item [WEB-1, WEB-2, WEB-4]; (c) human_aging/human_sexuality US institutional facts [DATASET-D9, D10, D11]; (d) potentially historical framings such as the Cawnpore item [DATASET-D18].

OC-3: INSUFFICIENT DOCUMENTATION — paper provides no Datasheet/Data Statement; only documents that questions were 'manually collected by graduate and undergraduate students' [Q19] without demographic breakdown. HF metadata indicates 'no-annotation' [DATASET Concern 8].

OC-4: Re-annotation by representative regional annotator pool (BAU, BRRI, PJTSAU, ANGRAU agronomists; Bangladeshi and Indian water-policy experts) would be required for any deployment-relevant subset; the paper offers no such pool.

OC-5: Aggregation methods are not described; the single-correct-answer MCQ format inherently erases minority perspectives [Q23], which is particularly concerning for cross-border and culturally-situated questions.

OC-6: The combined absence of regional annotator validation, the explicit US-normative framing of large subtasks, and the user-confirmed expectation of divergent correct answers constitute serious convergent and external validity violations.

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Q19	“The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books.” (p.3)
Q22	“Human-level accuracy on this test varies. Unspecialized humans from Amazon Mechanical Turk obtain 34.5% accuracy on this test. Meanwhile, expert-level performance can be far higher. For example, real-world test-taker human accuracy at the 95th percentile is around 87% for US Medical Licensing Examinations, and these questions make up our "Professional Medicine" task. If we take the 95th percentile human test-taker accuracy for exams that build up our test, and if we make an educated guess when such information is unavailable, we then estimate that expert-level accuracy is approximately 89.8%.” (p.3)
Q23	“we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions.” (p.3)

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D18	the peculiar aggravation of the Cawnpore massacres was this, that the deed was done by a subject race — by black men who dared to shed the blood of their masters… British journalist William Howard Russell, My Indian Mutiny Diary, 1860 — British colonial framing of Indian history
WEB-1	Active Bangladesh–India treaty renewal context — divergent correct answers likely (https://www.tbsnews.net/foreign-policy/bangladeshi-indian-engineers-monitor-water-flows-under-ganges-water-sharing-treaty)
WEB-2	Teesta dispute unresolved — Bangladesh and India would adjudicate water-allocation questions differently (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)
WEB-4	Teesta Master Plan with Chinese financing — Bangladeshi vs Indian framings differ (https://www.tbsnews.net/features/panorama/teesta-master-plan-and-longstanding-bangladesh-india-water-politics-1072696)
WEB-21	Kushiyara River Agreement — recent bilateral agreement not reflected in 2021 benchmark labels (https://www.waterdiplomat.org/story/2025/08/teesta-river-politics-and-benefit-sharing-getting-yes-without-grand-bargain)

Information Gaps

- Annotator demographic information is entirely absent from documentation
- No quality-assurance or inter-annotator agreement statistics reported

Requires Expert Verification

- Specific items where Bangladeshi vs Indian agronomic standards would diverge would require BAU/PJTSAU/ANGRAU expert review
- Cross-border water-policy item answer keys would require Joint Rivers Commission expert adjudication

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Remediation

Gap 1: No South Asian annotators were involved in label creation; user confirmed cross-border questions would yield divergent correct answers; HF metadata tags annotations as 'no-annotation'.

Recommendation: For any subset of MMLU items used in deployment evaluation, conduct a relabeling pass with at least three annotators per item drawn from a pool spanning Bangladesh (BAU, BRRI), Telangana (PJTSAU), and Andhra Pradesh (ANGRAU). Track inter-annotator disagreement rates and exclude items where regional disagreement exceeds threshold. Document annotator demographics in a Datasheet.